

Floorcast

From forecast to floor plan.

Akanchha Agarwal • WFM Simplified • July 2026

1. Objective

Floorcast answers a single question that today requires four disconnected tools and weeks of spreadsheet work: given a long-term volume forecast for each program and a floor of known square footage, what should the seating plan be — month by month, seat by seat, partition by partition?

The tool ingests three inputs — a 12-month forecast per program (volume, AHT, intraday profile), workforce parameters (service level targets, shrinkage, WFH share, shift rules), and a floor definition (area, dimensions, fixed obstacles, quiet zones, seat density norm). It outputs the complete downstream chain: required FTE per month, an optimized shift mix, interval-level seat demand, and a floor layout showing where movable partitions should be placed for each program’s zone — including how the layout should evolve as programs ramp.

2. From Objective to Implementation: The Five-Stage Pipeline

Floorcast chains five stages, each a well-defined computation, each independently verifiable:

- **Stage 1 — Forecast intake.** Monthly volumes are distributed across the day using intraday profiles and day-of-week mixes, producing contact demand per 30-minute interval.
- **Stage 2 — Capacity engine.** Erlang C converts interval demand into required agents-on-seat for the service level target, then grosses up for shrinkage to scheduled-headcount requirements.
- **Stage 3 — Shift-mix optimizer.** A set-covering integer program selects how many agents to place on each allowed shift template so every interval is covered with minimum total FTE — scheduling handled by construction, not assumption.
- **Stage 4 — Seat demand engine.** The shift mix, netted for WFH, becomes interval-level bodies-in-building concurrency per program; the peak defines each program’s seat requirement.
- **Stage 5 — Layout optimizer.** The floor is discretized into pods (6 seats ≈ 180 sq ft). A mixed-integer program assigns pods to programs — respecting quiet zones, building core, and client segregation — while minimizing total partition length, which is precisely the movable-division installation cost. Solved per month, it yields a partition evolution plan.

3. Comparison with Current Tools

No existing category covers this chain. Each stops one link short of the floor:

Capability	Workplace tools (Tango, Robin, Condeco)	WFM suites (NICE, Verint, Calabrio)	Floorcast
Forecast → FTE	No	Yes	Yes (Erlang C)
Shift-mix optimization	No	Yes (agent-level)	Yes (set covering)
Interval seat demand	Backward-looking utilization only	No — stops at schedules	Yes (overlap math, WFH-netted)
Floor layout / partitions	Manual drag-and-drop	No	Optimized; partition length minimized

Forecast-driven layout evolution	No	No	Yes — monthly re-plan with change list
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In practice, organisations bridge these gaps with Excel. The bridge is fragile: blanket seat-sharing ratios, manually maintained layouts, and no ability to answer “what does the floor look like in March if the ramp lands?” without days of rework.

4. Case Study: Meridian Services, Gurugram (Illustrative)

To demonstrate the pipeline end-to-end, we run a fictional but representative delivery center through Floorcast: a 14,400 sq ft floor (10 × 8 pod grid with a central building core and a designated quiet corner), three programs — CareVoice (24/7 voice, client-dedicated, ramping +60% over 12 months), ClaimsChat (chat with split shifts, 35% WFH), and BillingOps (business hours, 50% WFH) — at an 80/20 service level and 30% shrinkage. All figures below are actual solver outputs, not illustrations.

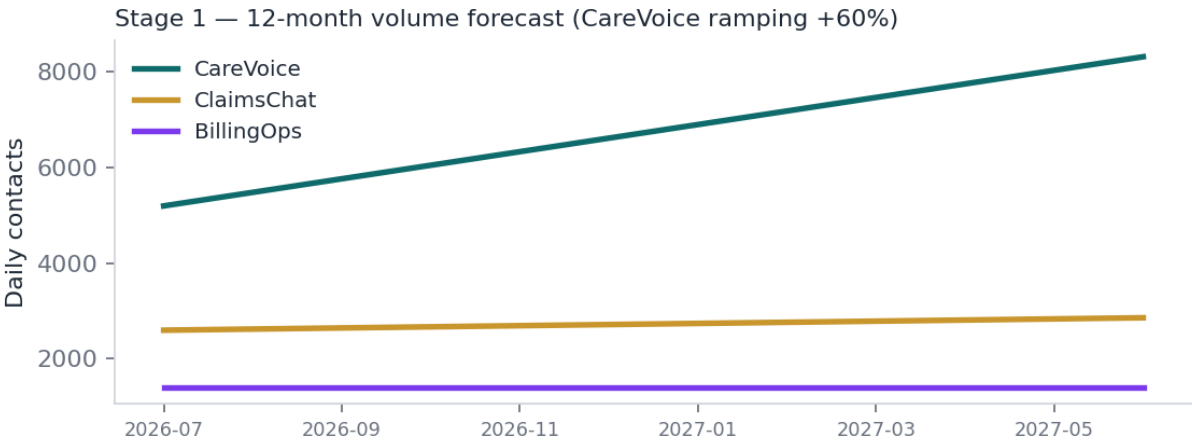


Fig 1. Stage 1 input: 12-month daily-volume forecast. CareVoice ramps from 5,200 to 8,320 contacts/day.

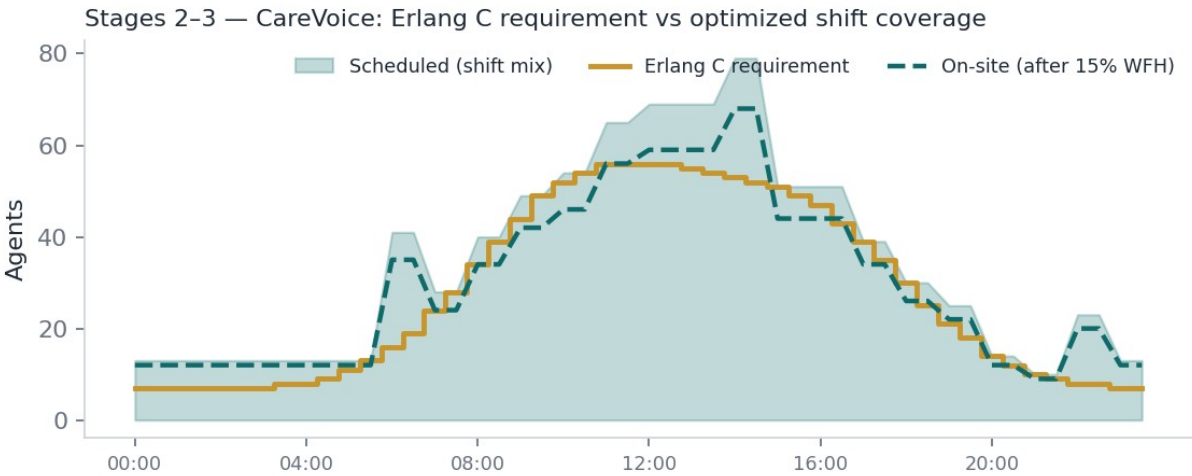


Fig 2. Stages 2-3: Erlang C interval requirement (gold) vs the optimized shift-mix coverage (teal area) and on-site line after WFH. The set-covering solver schedules 92 CareVoice agents across 7 shift templates.

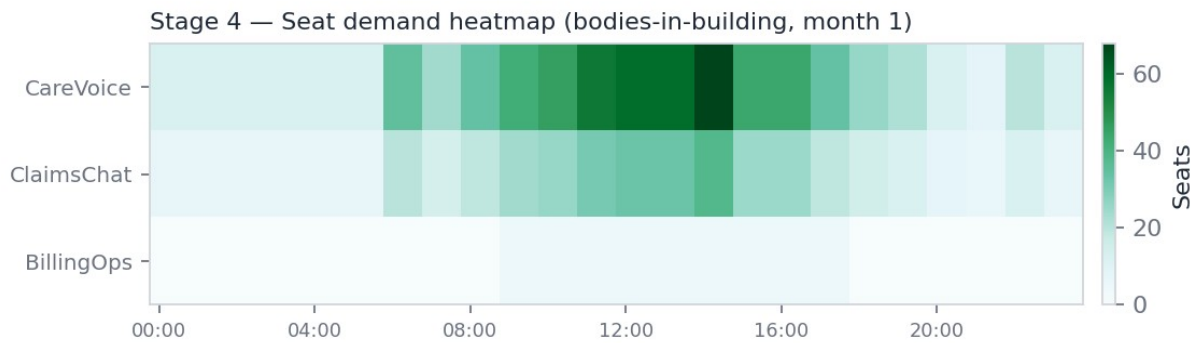


Fig 3. Stage 4: seat demand heatmap — bodies-in-building per 30-min interval. Peaks: CareVoice 68, ClaimsChat 38, BillingOps 5 — 111 seats total.

Stage 5 — Optimized floor layout: movable-partition plan (each pod = 6 seats ≈ 180 sq ft)

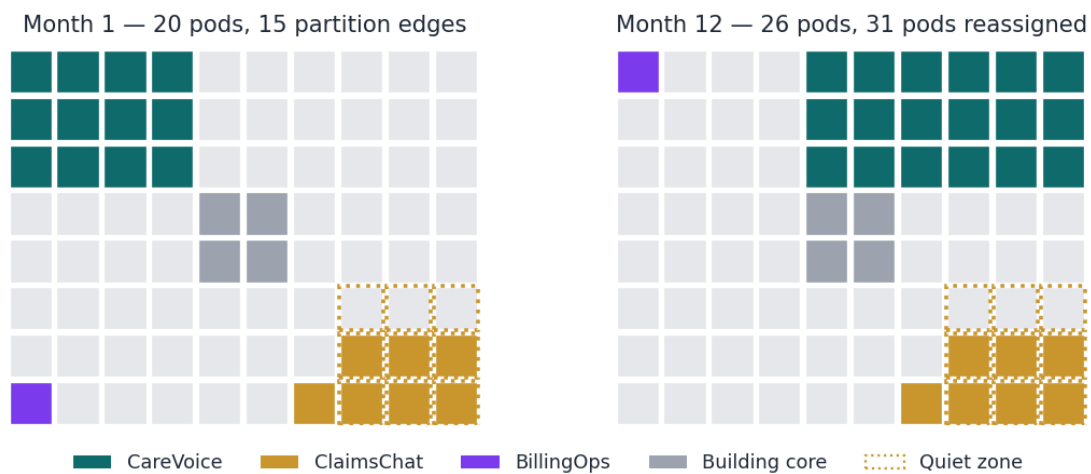


Fig 4. Stage 5: optimized pod layout. Month 1 uses 20 pods; by month 12 CareVoice grows from 12 to 18 pods and the partition plan updates accordingly, with the quiet corner (gold dotted) kept voice-free and the client-dedicated zone contiguous.

5. Cost Impact

The month-1 seat decision alone illustrates the money at stake. The two rule-of-thumb methods used across the industry both miss — in opposite directions:

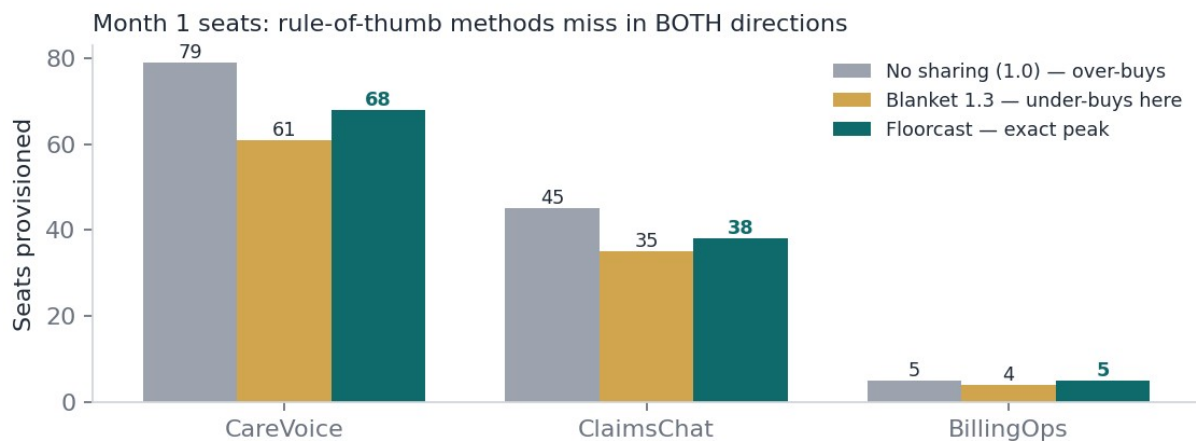


Fig 5. Seats provisioned under three methods. Rules of thumb err both ways; Floorcast provisions to the mathematically exact peak.

- **The safe default over-buys.** One-seat-per-office-agent provisions 129 seats where 111 suffice. At a fully loaded Grade-A NCR seat cost of ₹10,000/month (assumption, stated conservatively), those 18 excess seats cost ₹21.6 lakh per year on this one floor.
- **The blanket ratio under-buys.** A typical 1.3 sharing ratio provisions only 100 seats — leaving 11 agents without a seat at peak, a direct service-level incident that no facilities report would ever trace back to a planning assumption.
- **Partitions become a costed decision.** Because the layout objective minimizes boundary length, partition installation is minimized by construction; and the month-12 re-plan arrives as an explicit change list rather than a floor-wide reshuffle debate.
- **Planning cycle time collapses.** The forecast-to-floor chain that typically consumes days of cross-team spreadsheet work per re-plan runs in minutes, making monthly re-planning realistic instead of annual.

6. Roadmap and Boundaries

Phase 1 delivers Stages 1–4 (forecast to seat demand) as working modules; Phase 2 adds the layout optimizer and the monthly evolution view; Phase 3 extends the layout into workstation-level draft drawings — tiling individual desks within each program zone using configurable spacing and aisle norms, assigning seat numbers, and exporting to DXF so the output opens directly in AutoCAD as architect-ready input. Phase 4 adds layout stability (penalizing partition moves between months), Erlang A abandonment modeling, and multi-floor support.

A boundary stated without ambiguity: every layout Floorcast produces — zone plans and workstation-level drafts alike — is a planning draft, not a construction drawing. Fire and safety exits, egress travel distances, occupancy load limits, and accessibility clearances are statutory matters, and any proposed design must be counter-checked, corrected where required, and certified by a licensed architect specialising in fire-safety and egress design before implementation. Floorcast's role is to compress weeks of capacity-to-layout iteration into a strong first draft for that professional review — not to replace it. The reference implementation runs entirely on synthetic data.

Akanchha Agarwal is a workforce management professional with five years of experience in forecasting, capacity planning, and operations analytics across global contact center programs, and the creator of WFM Simplified. She holds an M.Tech from IIT Kanpur and a Lean Six Sigma Black Belt certification.